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Total No. of printed pages = 03

Monsoon, 2024

B.Tech 3rd Semester (Supplementary) Examination

Fluid Mechanics – Theory

Course Code: BCE23305T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

SECTION 1

Answer all the questions

Marks: 10 x 1 = 10

(a) Pascal-second is the unit of

(i) Pressure

(ii) Kinematic viscosity

☒ (iii) Dynamic viscosity

(iv) Surface tension

(b) Cippolletti weir has a side slope of

(i) 1: 2

(ii) 1.3

☒ (iii) 1.4

(iv) 1.5

(c) The intensity of pressure developed by surface tension of 0.075 N/m in a droplet of water of 0.075 mm diameter is

(i) 0.8 N/cm²

(ii) 0.6 N/ cm²

(iii) 0.2 N/ cm²

☒ (iv) 0.4 N/ cm²

(d) A floating body is said to be in state of stable equilibrium.

(i) When its metacentric height is zero.

☒ (ii) When the metacentre is above the centre of gravity

(iii) When the metacentre is below the centre of gravity

(iv) Only when its centre of gravity is below its centre of buoyancy

(e) Bernoulli's theorem deals with the principle of conservation of

☒ (i) energy

(ii) momentum

(iii) mass

(iv) force

- (f) Reynolds Number is ~~square root~~ of the ratio of
- (i) viscous force to gravity force
 - ☒ (ii) inertia force to viscous force
 - (iii) gravity force to viscous force
 - (iv) inertia force to gravity force
- (g) An open cubical tank of 2m side is filled with water. If the tank is rotated with an acceleration such that half of the water spills out, then the acceleration is equal to
- (i) $2g/3$
 - (ii) $g/3$
 - (iii) $g/2$
 - ☒ (iv) g
- (h) In MLT system the dimensions of mass density would be
- (i) L^3
 - (ii) ML^3
 - ☒ (iii) ML^{-3}
 - (iv) $M \cdot L^3$
- (i) 1 litre of crude oil weighs 9.6 N. Calculate its specific weight.
- ☒ (i) 9600 N/m^3
 - (ii) 978.6 kg/m^3
 - (iii) 0.978
 - (iv) 978 N/m^3
- (j) Venturimeter is a device used in the flowing fluid for measuring
- ☒ (i) discharge
 - (ii) pressure
 - (iii) velocity
 - (iv) kinetic energy

SECTION 2

Answer any one question from this section

Marks: 10

- 2 (a) The space between two square flat parallel plates is fitted with oil. Each side of the plate is 60 cm. The thickness of the oil film is 12.5 mm. The upper plate, which moves at 2.5 m/s requires a force 98.1 N to maintain the speed. Determine
- (i) The dynamic viscosity of the oil in poise
 - (ii) The kinematic viscosity of the oil in stokes if the specific gravity of the oil is 0.95
- (5+5=10)
- (b) Calculate the capillary rise in a glass tube of 2.5 mm diameter when immersed vertically in (i) water and (ii) mercury. Take surface tension $\sigma = 0.0725 \text{ N/m}$ for water and $\sigma = 0.52 \text{ N/m}$ for mercury in contact with air. Specific gravity for mercury is 13.6 and angle of contact is 130° .
- (5+5=10)

SECTION 3

Marks: 10

Answer any one question from this section

3 (a) A rectangular plane surface is 2m wide and 3m deep. It lies in vertical plane in water. Determine the total pressure and position of centre of pressure on the plane surface when its upper edge is horizontal and

(i) coincides with water surface

(ii) 2.5 m below the free water surface.

(10)

(b) The velocity potential function is given by $\phi = 5(x^2 - y^2)$. Calculate the velocity components at the point (4,5).

(10)

SECTION 4

Marks: 10

Answer any one question from this section

4 (a) A horizontal venturimeter with inlet and throat diameters 30 cm and 15 cm respectively is used to measure the flow of water. The reading of differential manometer connected to the inlet and throat is 20 cm of mercury. Determine the rate of flow. Take, $C_d = 0.98$.

(10)

(b) Derive the expression of maximum discharge over a triangular weir.

(10)

SECTION 5

Marks: 10

Answer any one question from this section

5 (a) Using Buckingham's Π theorem, show that the velocity through a circular orifice is given by $V = \sqrt{2gh} \Phi [D/H, \mu/(\rho V H)]$, where H is the head causing flow, D is the diameter of the orifice, μ is the coefficient of viscosity, ρ is the mass density and g is the acceleration due to gravity.

(10)

(b) Water is flowing through a pipe of diameter 30 cm at a velocity of 4 m/s. Find the velocity of oil flowing in another pipe of diameter 10 cm, if the condition of dynamic similarity is satisfied between the two pipes. The viscosity of water and oil is given as 0.01 poise and 0.025 poise. The specific gravity of oil = 0.8.

(10)